

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
df = pd.read_csv("Heart Disease data.csv")
```

```
# First five rows of dataset
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
0	52	1	0	125	212	0	1	168	0	1.0
1	53	1	0	140	203	1	0	155	1	3.1
2	70	1	0	145	174	0	1	125	1	2.6
3	61	1	0	148	203	0	1	161	0	0.0
4	62	0	0	138	294	1	1	106	0	1.9

	ca	thal	target
0	2	3	0
1	0	3	0
2	0	3	0
3	1	3	0
4	3	2	0

```
# Last five rows of dataset
df.tail()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang
1020	59	1	1	140	221	0	1	164	1
1021	60	1	0	125	258	0	0	141	1
1022	47	1	0	110	275	0	0	118	1
1023	50	0	0	110	254	0	0	159	0
1024	54	1	0	120	188	0	1	113	0

	slope	ca	thal	target
1020	2	0	2	1
1021	1	1	3	0
1022	1	1	2	0

```
1023      2      0      2      1
1024      1      1      3      0
```

```
#Important properties of values in dataset
df.describe()
```

	age	sex	cp	trestbps	chol
\count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000
std	9.072290	0.460373	1.029641	17.516718	51.59251
min	29.000000	0.000000	0.000000	94.000000	126.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000
75%	61.000000	1.000000	2.000000	140.000000	275.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000

	fbs	restecg	thalach	exang	oldpeak
\count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	0.149268	0.529756	149.114146	0.336585	1.071512
std	0.356527	0.527878	23.005724	0.472772	1.175053
min	0.000000	0.000000	71.000000	0.000000	0.000000
25%	0.000000	0.000000	132.000000	0.000000	0.000000
50%	0.000000	1.000000	152.000000	0.000000	0.800000
75%	0.000000	1.000000	166.000000	1.000000	1.800000
max	1.000000	2.000000	202.000000	1.000000	6.200000

	slope	ca	thal	target
count	1025.000000	1025.000000	1025.000000	1025.000000
mean	1.385366	0.754146	2.323902	0.513171
std	0.617755	1.030798	0.620660	0.500070
min	0.000000	0.000000	0.000000	0.000000
25%	1.000000	0.000000	2.000000	0.000000
50%	1.000000	0.000000	2.000000	1.000000

75%	2.000000	1.000000	3.000000	1.000000
max	2.000000	4.000000	3.000000	1.000000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1025 entries, 0 to 1024
```

```
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	age	1025 non-null	int64
1	sex	1025 non-null	int64
2	cp	1025 non-null	int64
3	trestbps	1025 non-null	int64
4	chol	1025 non-null	int64
5	fbs	1025 non-null	int64
6	restecg	1025 non-null	int64
7	thalach	1025 non-null	int64
8	exang	1025 non-null	int64
9	oldpeak	1025 non-null	float64
10	slope	1025 non-null	int64
11	ca	1025 non-null	int64
12	thal	1025 non-null	int64
13	target	1025 non-null	int64

```
dtypes: float64(1), int64(13)
```

```
memory usage: 112.2 KB
```

```
# Number of rows and columns
```

```
df.shape
```

```
(1025, 14)
```

```
#Name of columns
```

```
df.columns
```

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg',  
       'thalach',  
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],  
      dtype='object')
```

```
pd.isnull(df).sum
```

```
<bound method DataFrame.sum of          age    sex    cp  trestbps  
chol    fbs  restecg  thalach  exang  \  
0      False  False  False      False  False  False  False  False  
False  
1      False  False  False      False  False  False  False  False  
False  
2      False  False  False      False  False  False  False  False  
False  
3      False  False  False      False  False  False  False  False
```

```
False
4      False  False  False      False  False  False      False  False
False
...      ...      ...      ...      ...      ...      ...      ...
...
1020    False  False  False      False  False  False      False  False
False
1021    False  False  False      False  False  False      False  False
False
1022    False  False  False      False  False  False      False  False
False
1023    False  False  False      False  False  False      False  False
False
1024    False  False  False      False  False  False      False  False
False
```

	oldpeak	slope	ca	thal	target
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
1020	False	False	False	False	False
1021	False	False	False	False	False
1022	False	False	False	False	False
1023	False	False	False	False	False
1024	False	False	False	False	False

```
[1025 rows x 14 columns]>
```

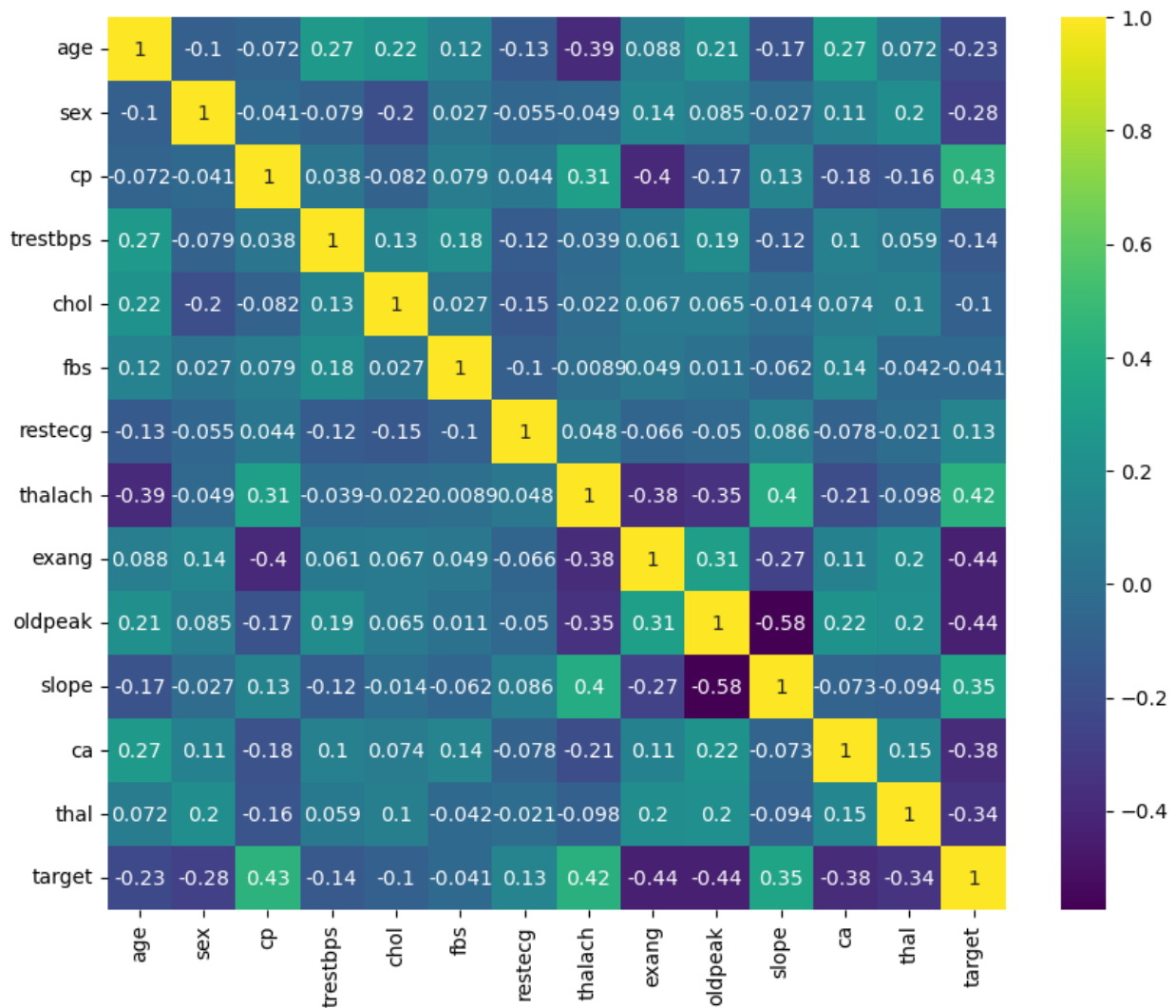
```
df.duplicated().value_counts()
```

```
True      723
False     302
Name: count, dtype: int64
```

```
# Correlations between different features of dataset
```

```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(),annot=True,cmap='viridis')
```

```
<Axes: >
```

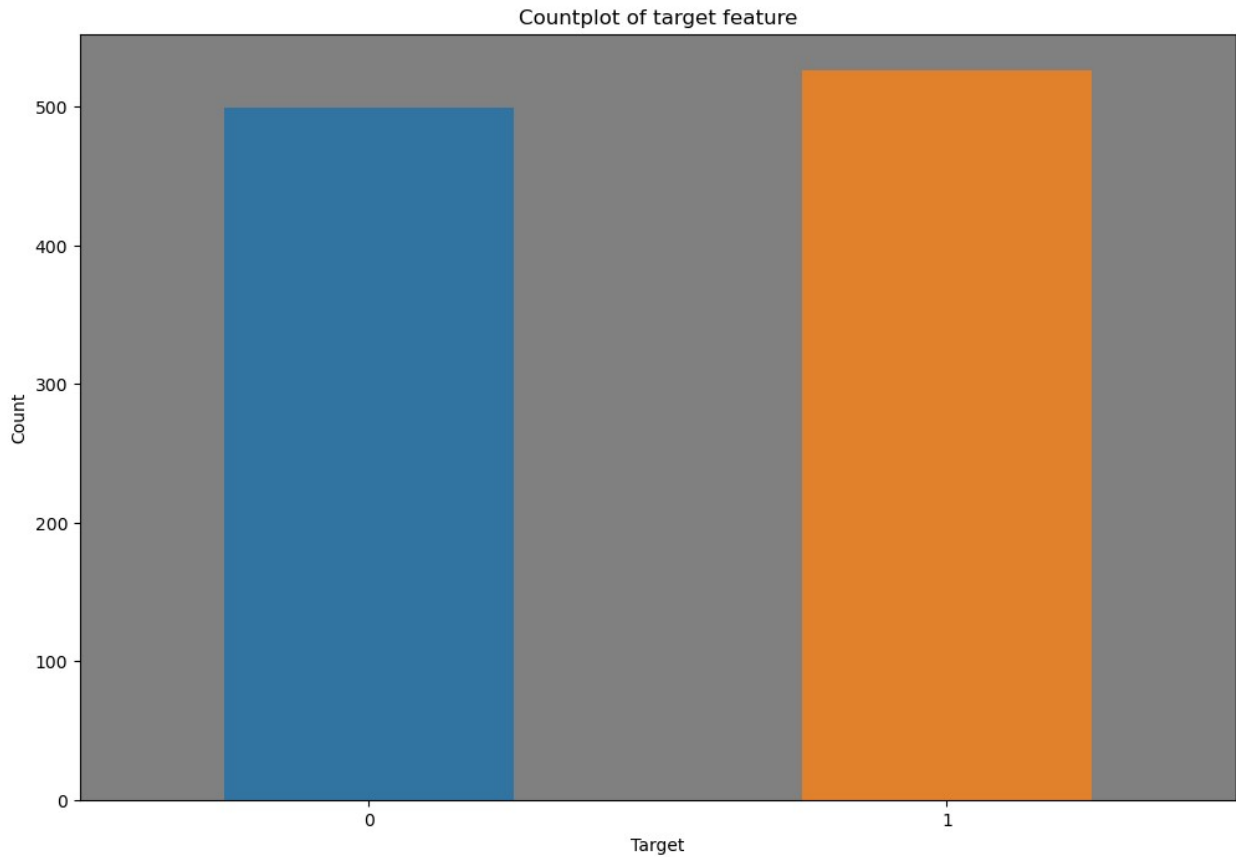


```
df['target'].value_counts()

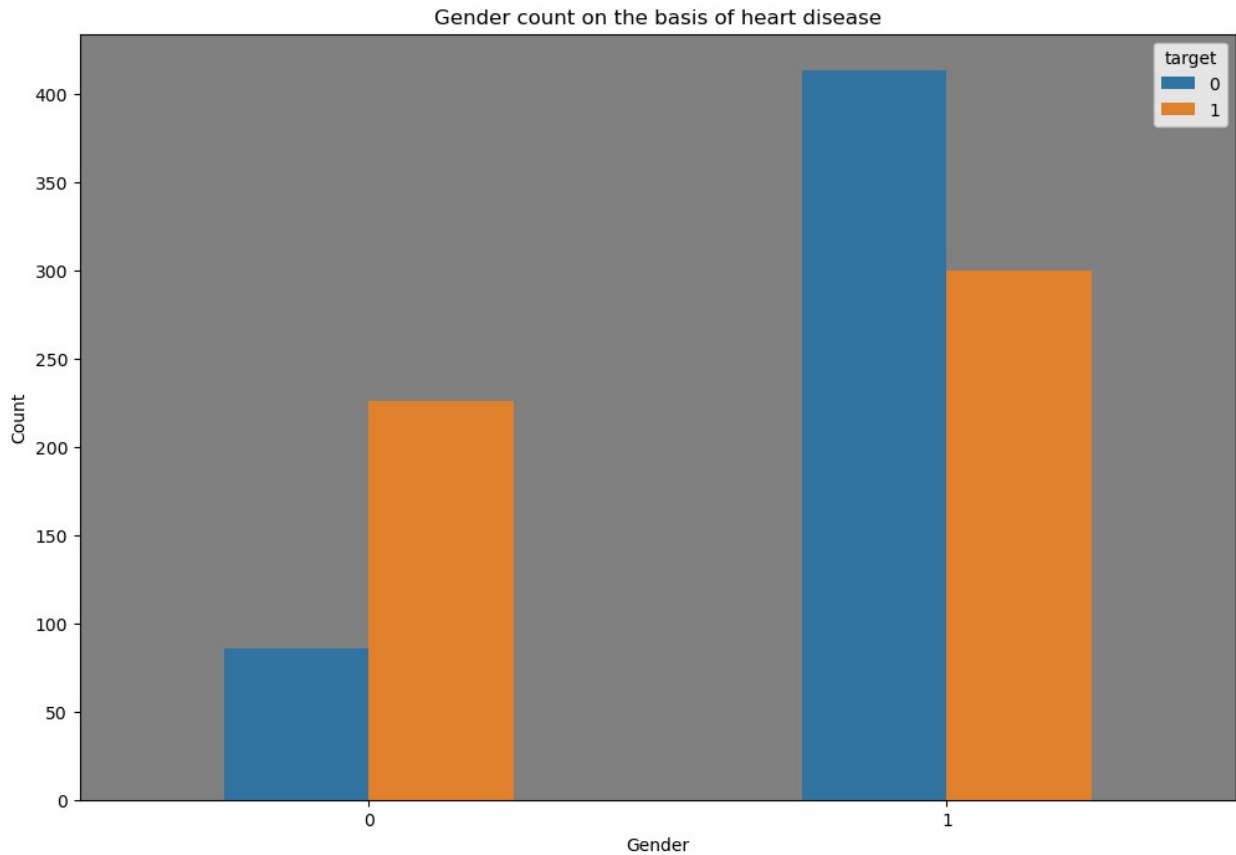
target
1    526
0    499
Name: count, dtype: int64

# To see visually through a countplot
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('grey')
sns.countplot(x='target',data=df,width=0.5)
plt.xlabel('Target')
plt.ylabel('Count')
plt.title('Countplot of target feature')

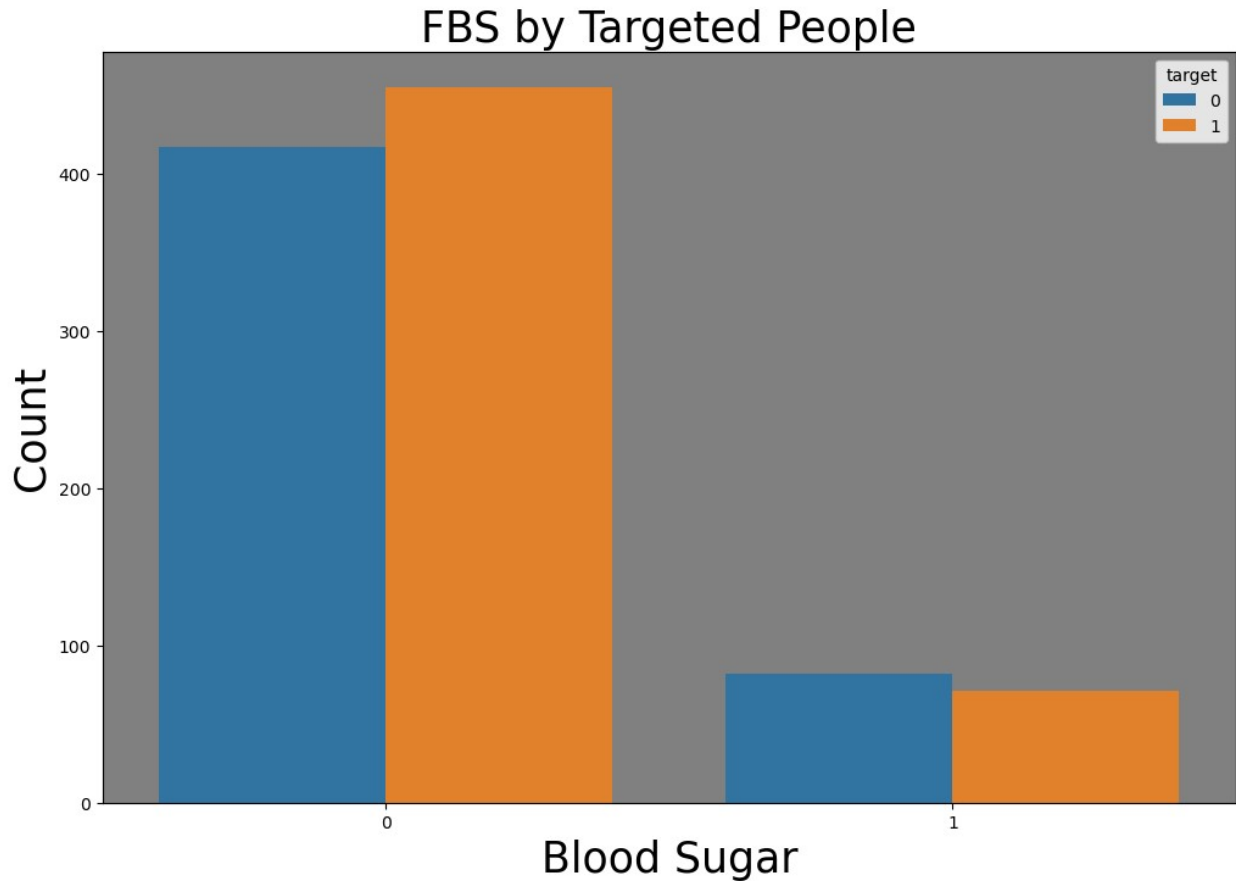
Text(0.5, 1.0, 'Countplot of target feature')
```



```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('grey')
sns.countplot(x='sex',hue='target',data=df,width=0.5)
plt.xlabel('Gender')
plt.ylabel('Count')
plt.title("Gender count on the basis of heart disease")
Text(0.5, 1.0, 'Gender count on the basis of heart disease')
```

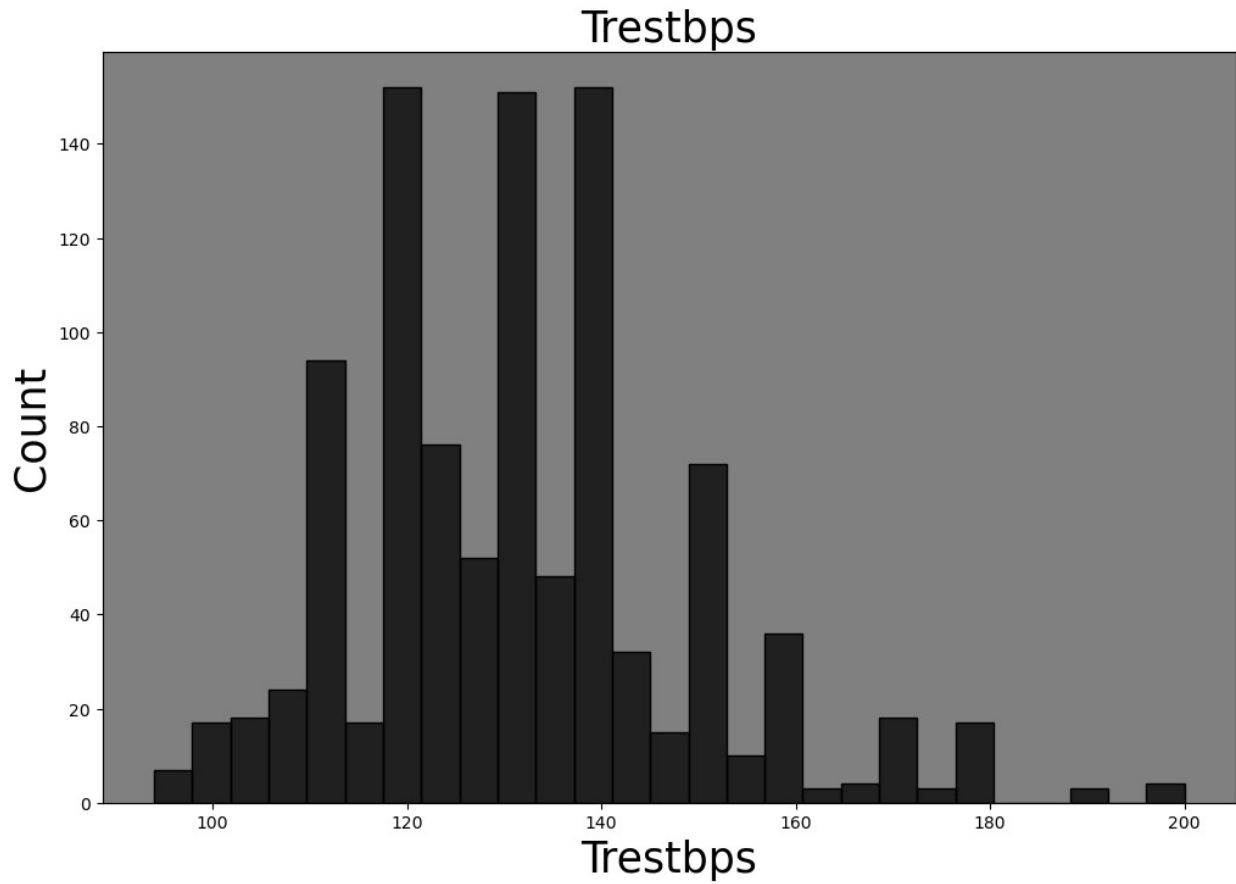


```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('grey')
sns.countplot(data=df,x='fbs',hue='target')# fasting blood sugar
distribution
plt.xlabel('Blood Sugar',fontsize=25)
plt.ylabel('Count',fontsize=25)
plt.title('FBS by Targeted People',fontsize=25)
Text(0.5, 1.0, 'FBS by Targeted People')
```



```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('grey')
sns.histplot(x='trestbps',data=df,color='black')
plt.show
plt.xlabel('Trestbps',fontsize=25)
plt.ylabel('Count',fontsize=25)
plt.title('Trestbps',fontsize=25)

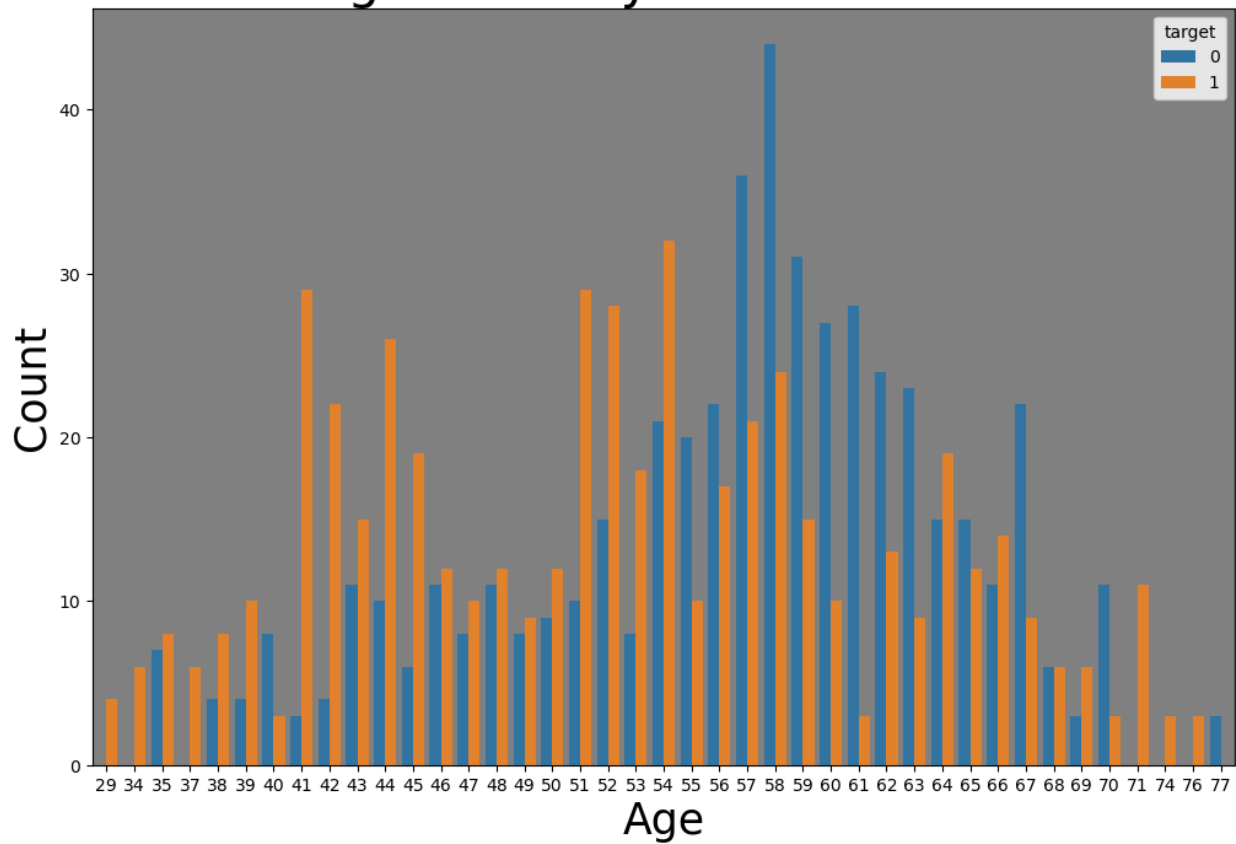
/Users/rs/anaconda3/lib/python3.11/site-packages/seaborn/
_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated
and will be removed in a future version. Convert inf values to NaN
before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
Text(0.5, 1.0, 'Trestbps')
```

```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('grey')
sns.countplot(x='age',data=df,hue='target')
plt.xlabel('Age',fontsize=25)
plt.ylabel('Count',fontsize=25)
plt.title('Age count by heart disease',fontsize=30)
plt.show

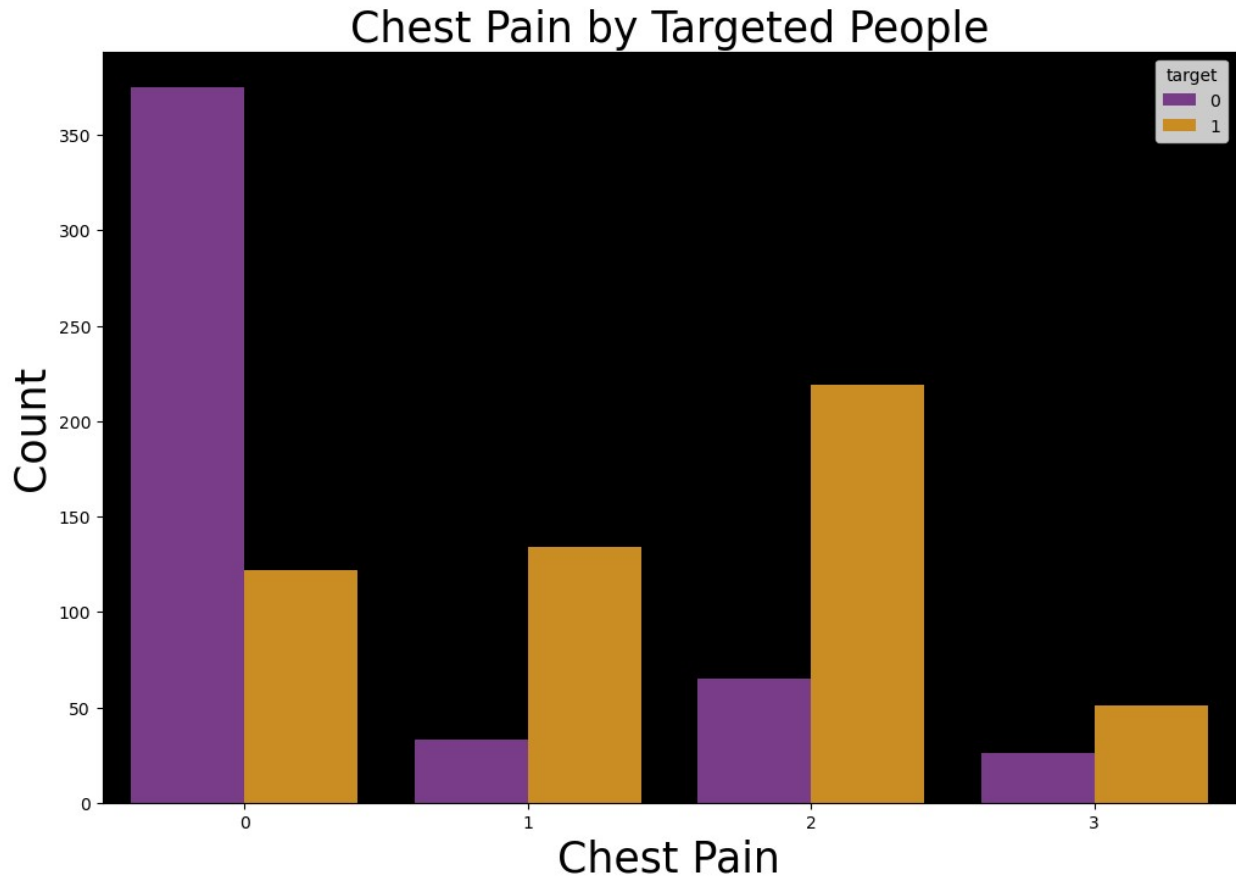
<function matplotlib.pyplot.show(close=None, block=None)>
```

Age count by heart disease



```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('black')
sns.countplot(data=df,x='cp',hue='target',palette='CMRmap')
plt.xlabel('Chest Pain',fontsize=25)
plt.ylabel('Count',fontsize=25)
plt.title('Chest Pain by Targeted People',fontsize=25)

Text(0.5, 1.0, 'Chest Pain by Targeted People')
```



```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('black')
sns.lineplot(x='cp',data=df,y='age',hue='sex')
#sns.set(rc={'figure.figsize':(15,5)})
plt.xlabel('Chest Pain',fontsize=25)
plt.ylabel('Age',fontsize=25)
plt.title('Chest Pain by Age',fontsize=25)
```

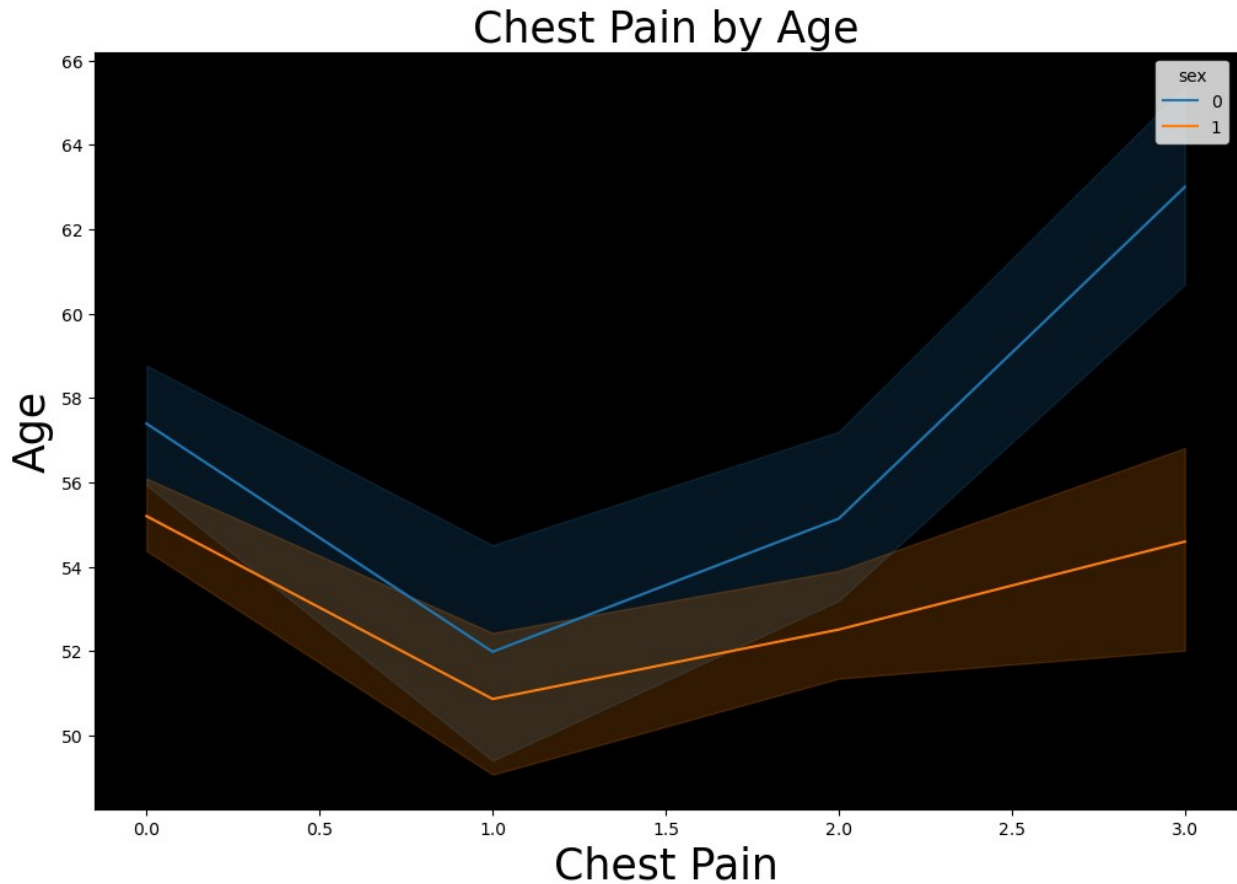
/Users/rs/anaconda3/lib/python3.11/site-packages/seaborn/_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

/Users/rs/anaconda3/lib/python3.11/site-packages/seaborn/_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

Text(0.5, 1.0, 'Chest Pain by Age')



```
plt.figure(figsize=(12,8))
ax = plt.axes()
ax.set_facecolor('black')
sns.histplot(x='chol',data=df,color='skyblue') # show distribution of
serum cholesterol
plt.xlabel('Cholesterol',fontsize=25)
plt.ylabel('Count',fontsize=25)
plt.title('Cholesterol',fontsize=25)
plt.show
```

```
/Users/rs/anaconda3/lib/python3.11/site-packages/seaborn/
_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated
and will be removed in a future version. Convert inf values to NaN
before operating instead.
```

```
with pd.option_context('mode.use_inf_as_na', True):
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```

